

$$\text{GP} : 2^0 + 2^1 + 2^2 + 2^3 + 2^4 = 2^5 - 1 < 2^5$$

$$\text{Sum of GP} \Rightarrow \frac{a(r^N - 1)}{r - 1}$$

$a \Rightarrow$  1st term

$N \Rightarrow$  no. of terms

$r \Rightarrow$  common ratio.

$$\rightarrow a = 1$$

$$N = 5$$

$$r = 2$$

$$\text{Sum} = \frac{1 \cdot (2^5 - 1)}{(2 - 1)} = 2^5 - 1$$

$$\text{GP} \Rightarrow 3^0 + 3^1 + 3^2 + 3^3 + 3^4 + 3^5 = \frac{3^6 - 1}{3 - 1} < \frac{3^6}{2}$$

$$\text{Sum} = \frac{1 \cdot (3^6 - 1)}{(3 - 1)} < \frac{3^6 - 1}{2}$$

$$a^0 + a^1 + a^2 + a^3 + \dots + a^{N-1} = \text{Sum} < \underline{a^N}$$

## Number systems

### 1) decimal no. system

digits  $\Rightarrow [0, 9]$

base  $\Rightarrow [10]$

$$\begin{array}{cccccc} 8 & 4 & 2 & 1 & 5 & \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \\ 10^4 & 10^3 & 10^2 & 10^1 & 10^0 & \end{array} \Rightarrow [8 \times 10^4 + 4 \times 10^3 + 2 \times 10^2 + 1 \times 10^1 + 5 \times 10^0]$$

$\begin{array}{cccccc} 5 & 4 & 3 & 2 & 1 & 0 \\ 6 & 4 & 3 & 2 & 1 & 5 \end{array}$

ii) binary no. system

digits  $\Rightarrow [0, 1]$

base  $\Rightarrow 2$ .

ex  $\Rightarrow$

|              |              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 5            | 4            | 3            | 2            | 1            | 0            |
| 1            | 0            | 1            | 1            | 0            | 1            |
| $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ |
| $2^5$        |              | $2^3$        | $2^2$        |              | $2^0$        |
|              | $2^4$        |              |              | $2^1$        |              |

$$= 1 \times 2^5 + \cancel{0 \times 2^4} + 1 \times 2^3 + 1 \times 2^2 + \cancel{0 \times 2^1} + 1 \times 2^0$$

$$= 2^5 + 2^3 + 2^2 + 2^0$$

$$= 32 + 8 + 4 + 1 = \underline{\underline{45}}$$

ex  $\Rightarrow$

|              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|
| 4            | 3            | 2            | 1            | 0            |
| 1            | 0            | 1            | 1            | 0            |
| $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ |
| $2^4$        | $2^3$        |              | $2^1$        | $2^0$        |
|              |              | $2^2$        |              |              |

$$2^4 + 2^2 + 2^1 = 16 + 4 + 2 = \underline{\underline{22}}$$

• decimal to binary:-

$$\begin{array}{r|l}
 2 & 45 \\
 \hline
 2 & 22 \\
 \hline
 2 & 11 \\
 \hline
 2 & 5 \\
 \hline
 2 & 2 \\
 \hline
 2 & 1 \\
 \hline
 & 0
 \end{array}
 \begin{array}{l}
 -1 \\
 -0 \\
 -1 \\
 -1 \\
 -0 \\
 -1
 \end{array}
 \begin{array}{l}
 \uparrow \\
 \uparrow \\
 \uparrow \\
 \uparrow \\
 \uparrow \\
 \uparrow
 \end{array}$$

1 0 1 1 0 1

$$\begin{array}{c}
 \downarrow \\
 32 + 8 + 4 + 1 = 45 \\
 \hline
 \checkmark
 \end{array}$$

$$\begin{array}{r|l}
 2 & 119 \\
 \hline
 2 & 59 \\
 \hline
 2 & 29 \\
 \hline
 2 & 14 \\
 \hline
 2 & 7 \\
 \hline
 2 & 3 \\
 \hline
 & 1 \\
 \hline
 & 0
 \end{array}
 \begin{array}{l}
 -1 \\
 -1 \\
 -0 \\
 -0 \\
 -1 \\
 -1 \\
 -1
 \end{array}
 \begin{array}{l}
 \uparrow \\
 \uparrow \\
 \uparrow \\
 \uparrow \\
 \uparrow \\
 \uparrow \\
 \uparrow
 \end{array}$$

$$\begin{array}{c}
 10011 \\
 \hline
 16 + 2 + 1 = 19 \\
 \hline
 \checkmark
 \end{array}$$

⇒ Adding two decimal nos:-

$$\begin{array}{r}
 \begin{array}{cccccc}
 & 1 & & 1 & & 1 \\
 & 6 & 4 & 5 & 9 & \\
 & 3 & 8 & 4 & 2 & \\
 \hline
 & 1 & 0 & 3 & 0 & 1
 \end{array}
 \end{array}$$

|               |       |       |       |
|---------------|-------|-------|-------|
| Sum = 11      | 10    | 13    | 10    |
| value = 11/10 | 10/10 | 13/10 | 10/10 |
| carry = 11/10 | 10/10 | 13/10 | 10/10 |

⇒ Adding two binary numbers:

ex ⇒

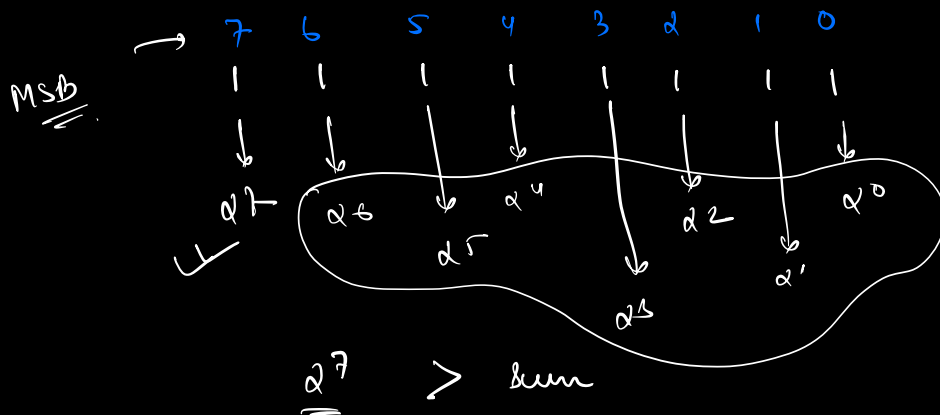
|               |                        |                        |                        |                        |                        |                        |    |
|---------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|----|
| 1             | 1                      | 1                      | 1                      | 0                      | 0                      |                        |    |
| 1             | 1                      | 0                      | 1                      | 0                      | 1                      | →                      | 53 |
| 0             | 1                      | 1                      | 1                      | 1                      | 0                      | →                      | 30 |
| 1 0 1 0 0 1 1 |                        |                        |                        |                        |                        | →                      | 83 |
| sum:          | 2                      | 3                      | 2                      | 2                      | 1                      | 1                      |    |
| value:        | $2 \times \frac{1}{2}$ | $3 \times \frac{1}{2}$ | $2 \times \frac{1}{2}$ | $2 \times \frac{1}{2}$ | $1 \times \frac{1}{2}$ | $1 \times \frac{1}{2}$ |    |
| carry:        | $\frac{2}{2}$          | $\frac{3}{2}$          | $\frac{2}{2}$          | $\frac{2}{2}$          | $\frac{1}{2}$          | $\frac{1}{2}$          |    |

∴ the number in binary:

10: 1 0 1 0

8 bit 10: 0 0 0 0 1 0 1 0

MSB ⇒ most significant bit



$$7^{th} \Rightarrow 2^7 \Rightarrow \underline{\underline{128}}$$

$$(0-6^{th}) \Rightarrow \underbrace{2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6}_{(2^7)} = \frac{1 \cdot (2^7 - 1)}{(2-1)}$$

$$= 2^7 - 1 \leftarrow \underline{\underline{2^7}}$$

$$10 \Rightarrow \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0$$

$$-10 \Rightarrow \quad \begin{array}{c} \diagdown \\ 1 \end{array} \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0$$

$$a \rightarrow -a \text{ [2's complement of } a \text{]}$$

↓

1's complement + 1

↓

flip (1  $\leftrightarrow$  0)

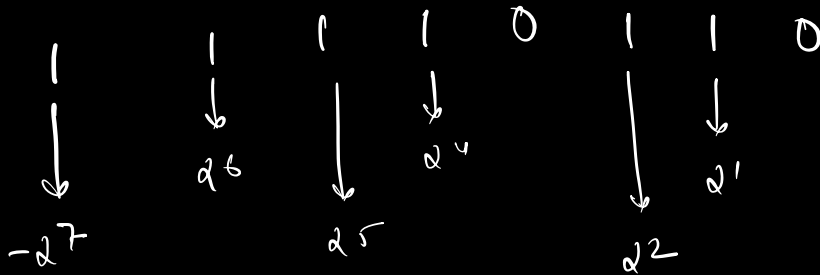
$$10 \Rightarrow \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0$$

$$1's \ 10 \Rightarrow \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1$$

$$+1 \Rightarrow \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1$$

$$\textcircled{-10} \rightarrow \quad \begin{array}{c} \hline 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 1 \quad 0 \\ \hline \end{array}$$

: MSB contribution is always negative.



$$\begin{aligned}
 & -2^7 + 2^6 + 2^5 + 2^4 + 2^2 + 2^1 \\
 & = -128 + 64 + 32 + 16 + 4 + 2 \\
 & = -128 + 118 = \underline{\underline{-10}}
 \end{aligned}$$

$$\begin{aligned}
 & 1101 \Rightarrow \underline{\underline{13}} \\
 & \downarrow \\
 & 8 \text{ bit} \mid 4 \text{ bit} \mid 16 \text{ bit}
 \end{aligned}$$

MSB

4 bit  $\Rightarrow$

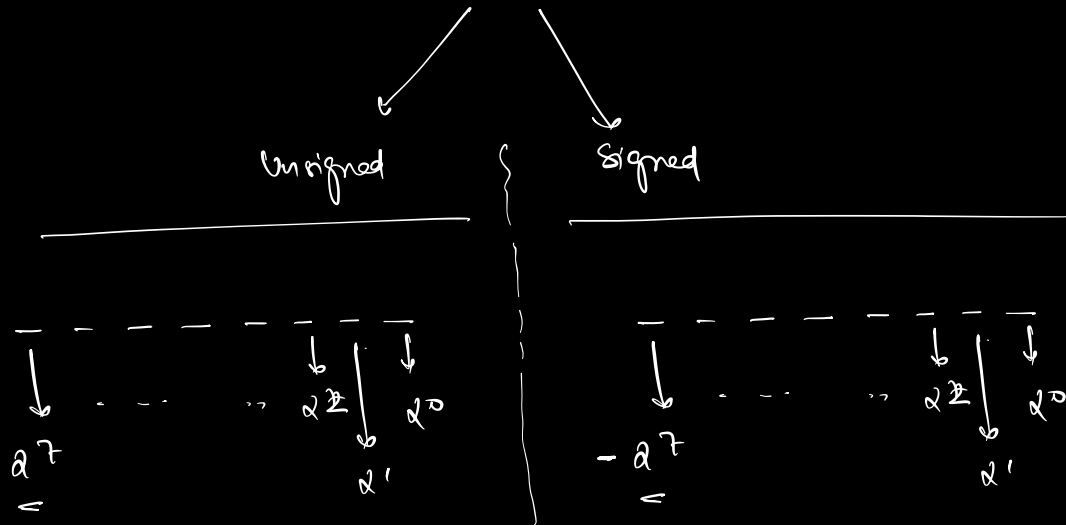
$$\begin{aligned}
 & 1101 \Rightarrow -2^3 + 2^2 + 2^0 \\
 & = -8 + 4 + 1 \\
 & = \underline{\underline{-3}}
 \end{aligned}$$

8 bits  $\Rightarrow$

$$\begin{aligned}
 & 00001101 \Rightarrow 2^3 + 2^2 + 2^0 \\
 & = 8 + 4 + 1 = \underline{\underline{13}}
 \end{aligned}$$

: Is MSB always negative?

Ans  $\Rightarrow$  depends | No.



|                     | unsigned                                                                                                                       | signed                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 8 bit               |                                                                                                                                |                                                                                                                                   |
| range $\Rightarrow$ | $\begin{array}{cccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \hline [0, 255] \end{array}$ | $\begin{array}{cccccccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \hline [-128, 127] \end{array}$ |

$\Rightarrow$  by default :-

$\left[ \begin{array}{l} \text{byte } n \\ \text{long } n \\ \text{int } n \end{array} \right] \downarrow \text{signed}$

$\left\{ \begin{array}{l} \text{unsigned int } n \\ \text{unsigned long } n \\ \text{unsigned byte } n \end{array} \right.$

C/C++/C#

## 11 Ranges

|                               | unsigned                                | signed                             |
|-------------------------------|-----------------------------------------|------------------------------------|
| 8 bits $\Rightarrow$          | $[0, 2^8 - 1]$<br>$[0, 255]$            | $[-2^7, 2^7 - 1]$<br>$[-128, 127]$ |
| 16 bits                       | $[0, 2^{16} - 1]$                       | $[-2^{15}, 2^{15} - 1]$            |
| N bits                        | $[0, 2^N - 1]$                          | $[-2^{N-1}, 2^{N-1} - 1]$          |
| integer $\Rightarrow$ 32 bits | $\longrightarrow [-2^{31}, 2^{31} - 1]$ |                                    |

$$\rightarrow 2^{10} = 1024 \Rightarrow 1000 \approx 10^3$$

$$\begin{aligned} & 2^{10} \approx 10^3 \\ & \hookrightarrow (2^{10})^3 \approx (10^3)^3 \end{aligned}$$

$$\Rightarrow 2^{30} \approx 10^9$$

$$\hookrightarrow 2 \times 2^{30} \approx 2 \times 10^9$$

$$\Rightarrow \boxed{2^{31} \approx 2 \times 10^9}$$

$\rightarrow$  integer range



## Bitwise operator :-

$\&$     $|$     $\wedge$     $\sim$     $<<$     $>>$   
 $\downarrow$     $\downarrow$     $\downarrow$     $\downarrow$     $\downarrow$     $\downarrow$   
 and   OR   XOR   negative   left shift   right shift

|   |   | $\&$ | $ $ | $\wedge$ | $\sim a$ |
|---|---|------|-----|----------|----------|
| a | b |      |     |          |          |
| 0 | 0 | 0    | 0   | 0        | 1        |
| 0 | 1 | 0    | 1   | 1        | 1        |
| 1 | 0 | 0    | 1   | 1        | 0        |
| 1 | 1 | 1    | 1   | 0        | 0        |

$\downarrow$   
 same same  
 puppy same

$a = 29$  :   0   0   0   1   1   1   0   1

$b = 18$  :   0   0   0   1   0   0   1   0

$a \& b$  :   0   0   0   1   0   0   0   0

$a | b$  :   0   0   0   1   1   1   1   1

$a \wedge b$  :   0   0   0   0   1   1   1   1

properties

$$\begin{array}{l|l}
 a \& b = b \& a & a \& b \& c = b \& a \& c = c \& a \& b \\
 a | b = b | a & a | b | c = b | a | c = c | a | b \\
 a \wedge b = b \wedge a & a \wedge b \wedge c = b \wedge a \wedge c = c \wedge a \wedge b
 \end{array}$$

$$a = 10 \text{ [decimal]}$$

$$a = 1010$$

$$1 = 0001$$

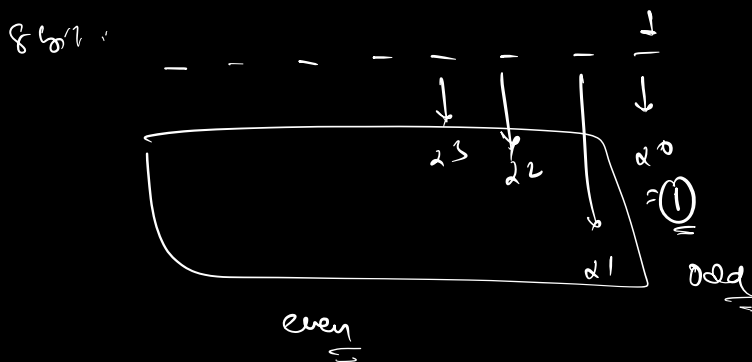
$$\hline a \& 1 = 0000$$

$$a = 11 \text{ [decimal]}$$

$$a = 1011$$

$$1 = 0001$$

$$\hline a \& 1 = 0001$$



if ( a & 1 == 1 ) {

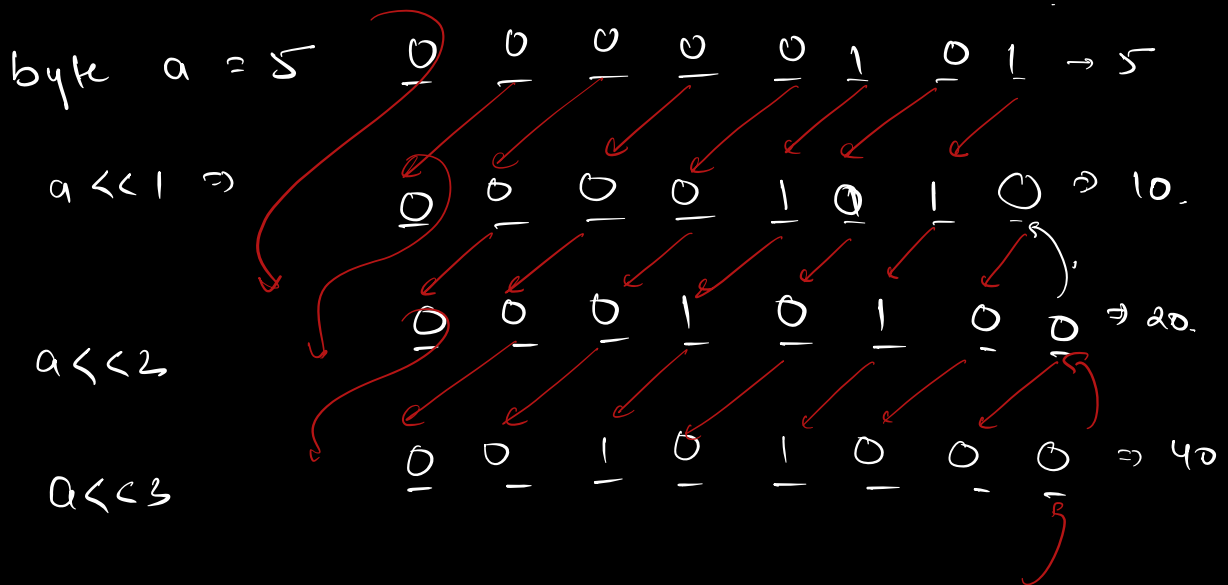
    a is odd

} else {

    a is even

}

## \* left shift and right shift :



$$a = 5$$

$$a \ll 1 \Rightarrow 5 \times 2 = 10$$

$$a \ll 2 \Rightarrow 5 \times 2^2 = 20$$

$$a \ll 3 \Rightarrow 5 \times 2^3 = 40$$

$\vdots$

$$a \ll N = 5 \times 2^N$$

$$a = 5$$

$$a \ll 5$$

$$5 \times 2^5 = 5 \times 32 = 160$$

$$\boxed{\begin{array}{l} a \ll N \Rightarrow a \times 2^N \\ 1 \leq N \leq 2^N \end{array}}$$

$\downarrow$   
until no overflow  
✓

$$\begin{array}{lcl}
 a = 50 & \Rightarrow & 0 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 0 \\
 a \gg 1 & \Rightarrow & 0 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \Rightarrow 25 \\
 a \gg 2 & \Rightarrow & 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0 \Rightarrow 12
 \end{array}$$

$$a \gg n \Rightarrow \frac{a}{2^n}$$

$$\frac{50}{2^1} = 25$$

$$\frac{50}{2^2} = 12$$

$$\frac{50}{2^3} = 6$$

$$\frac{50}{2^4} = 3$$

$$\frac{50}{2^5} = \frac{50}{32} = 1$$

$$\frac{50}{2^6} = \frac{50}{64} = 0$$

Q1. Given a no.  $N$ , check if  $i^{\text{th}}$  bit is set or not.

$$a \Rightarrow N = 29 \Rightarrow \begin{matrix} 4 & 3 & 2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{matrix}$$

$$i = 2$$

$$O/p \Rightarrow \text{true}$$

$$N = 29$$

$$i = 1$$

$$O/p = \text{false}$$

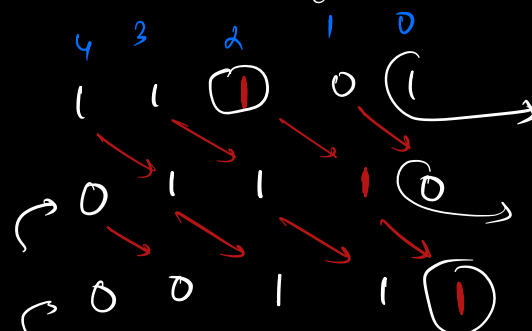
\* last bit

$$a \Rightarrow N = 29 \Rightarrow \begin{matrix} 4 & 3 & 2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{matrix}$$

$$i = \underline{0}$$

$$(a \& 1) == 1 \rightarrow \text{true}$$

$$== 0 \Rightarrow \text{false}$$

$$a \Rightarrow \underline{i = 2}$$


$$a \gg 1 \Rightarrow$$

$$\underline{a \gg 2} \Rightarrow$$

$$\left[ (a \gg 2) \& 1 \right] \rightarrow 1 \Rightarrow \text{return true}$$

$$\rightarrow 0 \Rightarrow \text{false}$$

give no.  $N$  bit position  $i$

$$\text{checkBit} \rightarrow \underline{\underline{[(N \gg i) \& 1]}} \Rightarrow \boxed{\begin{array}{l} TC \Rightarrow O(1) \\ SC \Rightarrow O(1) \end{array}}$$

Qd. Count no. of set bits in  $N$  (integer)

$$ex \Rightarrow N = 29 \Rightarrow 11101 \Rightarrow O/P = 4$$

$$N = 25 \Rightarrow 11001 \Rightarrow O/P = 3$$

$[0 \text{ to } 31] \rightarrow$  bits in integer.

$c = 0;$

for ( $i = 0; i \leq 31; i++$ ) {

if (checkBit( $N, i$ ))  
     $c++$

}

return  $c;$

$$\boxed{\begin{array}{l} TC \Rightarrow O(1) \\ SC \Rightarrow O(1) \end{array}}$$

$$\underline{\underline{a \wedge a}}$$

$$\boxed{\underline{\underline{a \wedge b}}}$$

$$\boxed{\underline{\underline{a \wedge b = x}}}$$

